Proceedings of the International Conference on Sustainable Communication Networks and Application (ICSCNA-2024)
IEEE Xplore Part Number: CFP24DW8-ART; ISBN: 979-8-3315-3001-3

SmartSignQuiz
A Colour-based Interactive Quiz System for Hard to Talk and Hard to Hear Students

Dr. N. Subbulakshmi
Associate Professor, Department of Computer Science and Engineering
Kalasalingam Academy of Research and Education
Krishnan Koil, Tamilnadu, India
n.subbulakshmi@klu.ac.in

Dr. R.Chandru
Assistant Professor, Department of Electronics and Communication Engineering
Sri Ramakirshna Engineering College,
Tamilnadu, India
r.chandru@srec.ac.in

Dr. S. Ariffa Begum
Associate Professor, Department of Computer Science and Engineering
Kalasalingam Academy of Research and Education
Krishnan Koil, Tamilnadu, India
s.ariffabegam@klu.ac.in

Manju Vallabha Pavalla
Department of Computer Science and Engineering
Kalasalingam Academy of Research and Education
Krishnan Koil, Tamilnadu, India
99210041261@klu.ac.in

Mohammed Muzammil Palagiri
Department of Computer Science and Engineering
Kalasalingam Academy of Research and Education
Krishnan Koil, Tamilnadu, India
9921004524@klu.ac.in

2024 International Conference on Sustainable Communication Networks and Application (ICSCNA) | 979-8-3315-3001-3/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICSCNA63714.2024.10864242

***Abstract*— The growing field of artificial intelligence (AI) provides new opportunities to develop inclusive educational tools tailored to diverse student needs. This study introduces SmartSignQuiz, an innovative colour-based interactive quiz system for hard-to-talk and hard-to-hear students. The system combines facial recognition and colour detection technologies, facilitating real-time participation without relying on verbal communication. SmartSignQuiz ensures accurate student identification through a face recognition module while employing colour-coded responses for quiz interaction. The primary objective is to create an accessible, non-verbal learning platform that promotes active engagement for all students. The study evaluates the system's accuracy, computational efficiency, and impact on educational outcomes, demonstrating its potential to enhance inclusivity in classroom environments.**

***Keywords*—** *Facial Recognition, Colour Detection, Computer Vision*

I. INTRODUCTION

In an increasingly digital world, the inclusion of students with diverse needs remains a critical challenge in educational technology. Among these needs, ensuring effective learning tools for deaf and hard-of-hearing students is of paramount importance. Traditional educational methods and resources often fall short in providing engaging and accessible content for these learners, particularly when it comes to interactive and visually-driven activities. These limitations can hinder their academic progress and limit their ability to engage with the curriculum on an equal footing with their peers.

Deaf and hard-of-hearing students face unique challenges in educational settings, relying heavily on visual and tactile information to facilitate learning. Classroom environments often assume the ability to hear spoken language as a primary mode of instruction, which can create barriers for students who depend on sign language, lip-reading, and other visual cues. While sign language interpretation, captioning services, and other accommodations are beneficial, they do not always offer an interactive or immersive learning experience. Furthermore, there is a lack of tailored educational tools that specifically address the interactive learning styles of these students.

Interactive quizzes are a crucial element of modern education, aiding in the reinforcement of concepts and providing immediate feedback to learners. However, most existing quiz systems are designed with an audio-visual approach that may not be fully accessible to deaf and hard-of-hearing students. There is a pressing need for a system that transcends these barriers and provides an inclusive, engaging, and effective learning environment. This necessitates the development of educational technologies that can cater to the unique needs of these students while enhancing their overall learning experience.

To address this gap, the development of innovative systems that leverage advanced technologies is essential. This paper introduces **SmartSignQuiz**, a novel colour-based interactive quiz system designed specifically for deaf and hard-of-hearing students. SmartSignQuiz integrates state-of-the-art AI-driven facial and gesture recognition technologies to create an engaging and adaptive learning experience. By utilizing these technologies, the system is able to interpret users' responses through gestures and facial expressions, providing a natural and intuitive mode of interaction that aligns with the communication styles of deaf and hard-of-hearing individuals.

Authorized licensed use limited to: University of New South Wales. Downloaded on October 22,2025 at 13:45:08 UTC from IEEE Xplore. Restrictions apply.

The design of SmartSignQuiz revolves around the innovative use of colour coding, which serves as both a navigational aid and a means of reinforcing learning objectives. The colour-based interaction model not only simplifies the interface, making it more intuitive, but also enhances visual engagement. This method provides an alternative to text-heavy interfaces, allowing students to interact with content in a manner that is visually stimulating and easily comprehensible. Furthermore, colour-coded feedback mechanisms are employed to indicate correct and incorrect responses, providing immediate and clear feedback that is essential for effective learning.

The implementation of SmartSignQuiz represents a significant advancement in the field of educational technology for special needs. Its unique combination of AI-driven facial and gesture recognition with a colour-based interactive framework offers a holistic solution that not only addresses accessibility concerns but also enriches the learning process. By enabling deaf and hard-of-hearing students to interact with quiz content in a manner that resonates with their natural modes of communication, SmartSignQuiz has the potential to significantly enhance their educational outcomes.

This paper explores the design, development, and implementation of SmartSignQuiz, detailing the technical aspects of its AI-driven features and the colour-based interaction model. We examine the potential of SmartSignQuiz to enhance the learning experience for deaf and hard-of-hearing students, highlighting both its innovative approach and its practical applications in educational settings. The system is evaluated in terms of usability, effectiveness, and impact on student engagement and learning outcomes.

Through this research, we aim to contribute valuable insights into the creation of effective, inclusive educational technologies and to set a precedent for future innovations in this field.

II. LITERATURE SURVEY

[1] Joseph Bamidele Awotunde et al. (2024). This paper investigates the use of the K-Nearest Neighbour (KNN) classifier for detecting multiple colours in RGB images. The researchers utilized a colour histogram for feature extraction to identify the most relevant patterns defining specific colours. By employing the KNN algorithm, a popular machine learning classification technique, they were able to effectively segregate distinct colours in images. The study's findings suggest that this approach enhances the accuracy and adaptability of colour classification, especially in applications involving complex image datasets. The proposed method demonstrates potential for various image processing tasks requiring precise colour identification.

[2] J. Sainath et al. (2024). This paper explores the integration of deep learning techniques with OpenCV and Pillow libraries for advanced colour detection. The authors emphasize the use of clustering algorithms, particularly for colour quantization and segmentation. By leveraging deep learning frameworks, they were able to analyze image data to extract dominant colour information more robustly. The study also discusses implementation details and evaluation metrics, offering insights into future advancements in real-world colour detection applications. The proposed methodology enhances detection accuracy, making it suitable for applications such as object recognition and image editing.

[3] Aadarsh Nayyer et al. (2024). The study introduces an innovative approach to object recognition by integrating colour detection and machine learning techniques. Utilizing YOLO v3, a state-of-the-art object detection algorithm, combined with k-means clustering, the authors were able to improve the accuracy and efficiency of object recognition tasks. The integration of colour and shape features demonstrated superior performance compared to traditional methods using only colour features. The study's results are particularly impactful for applications in robotics, autonomous vehicles, and security systems, where accurate object recognition is crucial.

[4] Patil Siddhant Vitthal et al. (2024).This paper addresses the challenge of colour analysis in textile industries using image processing techniques. The authors employed a GigE camera to capture fabric images and used k-means clustering and Euclidean distance methods for colour matching. Their experiments showed that k-means clustering provided better results than the Euclidean distance method due to its ability to identify different colour clusters effectively. The study highlights the importance of automated colour analysis in improving the quality and efficiency of fabric inspection, reducing reliance on manual assessment, which can be costly and error-prone.

[5] Yuko Toyokawa et al. (2024).This research focuses on the role of artificial intelligence (AI) in promoting inclusive education, particularly in the context of special needs students. The study employs the Learning and Evidence Analysis Framework (LEAF) to explore the application of AI in a learning analytics-enhanced environment. Through the Active Reading task, the authors analyze learning behaviours and assess the challenges and future potential of using AI technology to cater to diverse learner needs. The findings suggest that AI can significantly contribute to creating an inclusive learning environment by tailoring educational experiences to individual student requirements.

[6] Hajrah Sultan et al. (2024). This paper presents a real-time face recognition-based attendance system for university classrooms. Utilizing the Dlib API and facial recognition technology, the system captures and matches student faces to automate attendance recording. The proposed system achieved a high accuracy rate, demonstrating its effectiveness in reducing manual attendance errors and enhancing classroom management efficiency. This study emphasizes the potential of facial recognition technology in educational settings to streamline administrative processes and provide a more seamless experience for both educators and students.

[7] Ishteyaaq Ahmad et al. (2024). The study explores how Industry 4.0 technologies, including AI and IoT, can support inclusive learning environments by addressing the diverse needs of students. The authors highlight the potential of adaptive learning systems, enhanced by AI, to provide

Authorized licensed use limited to: University of New South Wales. Downloaded on October 22,2025 at 13:45:08 UTC from IEEE Xplore. Restrictions apply.

personalized educational experiences. This research discusses various Industry 4.0 applications, such as intelligent tutoring systems and sensor-based feedback, which can dynamically adjust to different learning styles and abilities. The study suggests that embracing these technologies can lead to more equitable and accessible educational outcomes.

[8] Rustam Abdrakhmanov et al. (2024). This paper discusses the application of computer vision and machine learning technologies to enhance self-study in STEM education. The authors propose using these technologies to improve student engagement by enabling interactive and hands-on learning experiences. Their system captures student interactions and analyzes them to provide feedback and adapt educational content. The findings indicate that such an approach can significantly enhance learning outcomes, offering a more immersive and effective self-study environment for STEM students.

[9] Dhaya Sindhu Battina et al. (2024). This research introduces an AI voice-based device designed to assist deaf individuals in interpreting and responding to body language. The system integrates gesture recognition with speech synthesis to facilitate non-verbal communication. By recognizing body language cues and providing audio feedback, the device aims to bridge the communication gap for deaf individuals. The study demonstrates the potential of AI and machine learning in creating assistive technologies that promote inclusivity and enhance the daily interactions of individuals with hearing impairments.

[10] Chandan Kumar Dubey (2024). This paper examines the potential of AI in fostering inclusive education. It focuses on adaptive learning technologies that cater to individual student needs, ensuring that diverse learning styles are accommodated. The author also discusses the ethical considerations involved in implementing AI-driven educational environments, emphasizing the need for responsible AI deployment. The study highlights the transformative role of AI in creating more accessible and personalized educational experiences, advocating for its integration into modern educational practices.

[11] This research focuses on deep learning-based hand gesture recognition to improve human-machine interaction. By employing a combination of skin color modeling, AdaBoost classifiers, and convolutional neural networks (CNNs), the study achieved 98.3% accuracy in recognizing common hand gestures. The results emphasize the significance of CNNs in accurately interpreting gestures in real time, particularly for robotics and intelligent systems.

[12] This paper introduces an AI-based mouse system using face recognition and hand gesture recognition for Human-Computer Interaction (HCI). The system integrates Eigenface algorithms for face authentication with deep learning models for hand gesture recognition, allowing users to control mouse functions without hardware. This study illustrates the practicality of computer vision in everyday tasks, emphasizing the importance of accessible and intuitive HCI.

III. METHODOLOGY

In the SmartSignQuiz system, computer vision plays a central role, utilizing several key techniques to facilitate facial recognition and color detection. The complexity of the proposed model lies in its real-time processing capabilities, which involve facial recognition and color detection modules. The system uses OpenCV and the face_recognition library, both of which are optimized for efficient video frame processing. The deep learning model for facial recognition has a computational cost proportional to the number of faces in the database, requiring high processing power for larger datasets. However, the system maintains real-time performance with modest hardware by limiting image resolution and optimizing data storage with lightweight encoding files. The color detection module, using HSV color space analysis, has a minimal computational footprint, enabling quick recognition even in changing lighting conditions. This library uses a deep learning-based approach for face detection and recognition, providing a highly accurate method to identify students in real-time.

In the system, computer vision techniques are applied in two main areas: face recognition and color detection. For facial recognition, the process begins with capturing video frames from a webcam using OpenCV. Each frame is resized and converted from BGR (the default color format in OpenCV) to RGB to align with the input requirements of the face_recognition library. The face_recognition library simplifies the complex process of face detection and recognition using deep learning. Initially, it detects faces in an image using a deep neural network that has been trained on a large dataset of face images. This network identifies key facial features, such as the eyes, nose, and mouth, to determine the precise location of the face within the image. After detecting the face, the library extracts facial encodings by mapping the detected features onto a 128-dimensional space. These encodings are essentially feature vectors that encapsulate the unique characteristics of a person's face, making it possible to distinguish between different individuals.

When the system needs to identify a face, it computes the face encoding for the detected face in the real-time video frame. It then compares this encoding to the set of known encodings stored during the initial data collection phase. The comparison is done using the Euclidean distance metric, which measures how close the new encoding is to the known encodings. A lower distance indicates a higher likelihood of a match. If the computed distance is below a certain threshold, the system recognizes the face as belonging to one of the known students. This method allows for highly accurate recognition even in varying lighting conditions and angles, making it ideal for real-world educational environments.

This section introduces the SmartSignQuiz system, outlining its purpose as an interactive quiz platform designed specifically for deaf and hard-of-hearing students. It highlights the integration of facial recognition, gesture recognition, and color detection technologies to facilitate an inclusive learning experience.

Authorized licensed use limited to: University of New South Wales. Downloaded on October 22,2025 at 13:45:08 UTC from IEEE Xplore. Restrictions apply.

A. Data Collection and Face Encoding

The system starts by gathering facial images of students, which are stored in a designated folder with filenames acting as unique student identifiers. Using OpenCV and the face_recognition library, the system extracts facial encodings from these images. This involves converting images to the RGB format and detecting facial landmarks to generate a unique numerical representation for each face. These encodings are then saved in a file ('EncodeFile.p') using Python's pickle module, serving as a reference for student identification during quiz sessions. The evaluation of SmartSignQuiz utilized a dataset of facial images collected from participating students, which is stored securely for privacy reasons. These images were processed using OpenCV and face_recognition libraries to generate unique facial encodings. The dataset references have been included to ensure transparency and enable reproducibility of the research findings. Future studies will expand the dataset to include diverse demographic groups, enhancing the generalizability of the results.

B. Real-Time Face Recognition and Identification

The system employs a webcam to capture real-time video frames. OpenCV is used to initialize video capture, ensuring that the resolution is adequate for precise facial feature capture. The face_recognition library detects faces in each video frame, computes their encodings, and compares them to the stored encodings to identify the student. Functions like compare faces and face distance are used to match the real-time encodings with known faces, enabling accurate and swift student identification.

C. Color-Based Interaction for Quiz Responses

The Color-Based Interaction for Quiz Responses module allows students to interact with the quiz system using colored cards. Upon recognizing a student, the system identifies a specific region of interest (ROI) in the video frame where the student is expected to hold a colored card, typically near their face or hands. The ROI is converted to the HSV color space, enhancing color detection accuracy by accounting for different lighting conditions. Predefined HSV ranges help identify specific colors like red, green, and blue, which serve as the students' method of answering quiz questions.

D. Region of Interest (ROI)

After successfully identifying a student in the frame, the system dynamically defines a Region of Interest (ROI) where the color card is expected to be held. Typically, this ROI is positioned near the student's body, such as the area to the bottom right or left of the detected face. The location of the ROI can be adjusted based on practical usage, ensuring that the color card is within the camera's view while minimizing interference from the background.

E. Color Detection Mechanism for Response Identification

The system extracts the Region of Interest (ROI) from the video frame based on predefined coordinates relative to the student's face to focus on the area where the color card is expected to be held. To detect colors accurately within this ROI, the system converts the image from the BGR color space (used by OpenCV) to the HSV (Hue, Saturation, Value) color space. The HSV color space is more effective for color detection as it separates the chromatic content (color) from intensity, making it more resilient to lighting variations. The system then uses predefined HSV ranges for the target colors, such as red, green, and blue. For each color, it applies a mask to the ROI using these HSV ranges, isolating pixels that fall within the specified range. The color with the highest number of matching pixels in the ROI is identified as the detected color. For instance, if most pixels in the ROI fall within the HSV range for blue, the system classifies the color as "Blue."

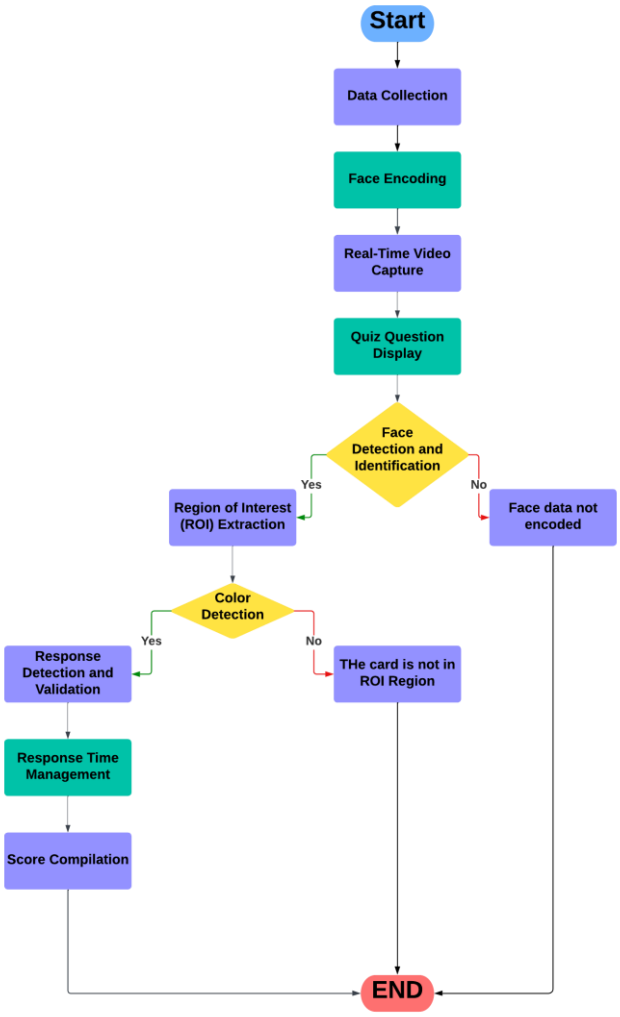


Fig 1 : Flowchart of the Proposed System

F. Quiz Execution and Response Tracking

During the Quiz Execution and Response Tracking phase, the system manages the quiz flow. It displays multiple-choice questions, each linked to a specific color response. As students hold up colored cards, the system continuously monitors the ROI for changes. The detected color is then compared against the correct answer for the question. If there is a match, the student's score is incremented. The system also tracks the time taken for each student to respond, logging "No response" if a student does not answer within the specified time limit, typically 10

Authorized licensed use limited to: University of New South Wales. Downloaded on October 22,2025 at 13:45:08 UTC from IEEE Xplore. Restrictions apply.

seconds. The quiz advances to the next question either after receiving all responses or reaching the time limit, repeating this process until all questions are completed.

G. Final Score Calculation and Output

After the quiz, the system enters the Final Score Calculation and Output phase, where it compiles scores for each student, providing a summary of correct answers. This final scoring gives educators a clear assessment of student performance. Additionally, the system can store responses and scores for future analysis, helping educators identify areas where students may need additional support or adaptive teaching methods.

The SmartSignQuiz system employs facial recognition and color detection to facilitate an interactive quiz experience for deaf and hard-of-hearing students. Initially, facial images are collected and encoded to create a database of student profiles. During the quiz, the system captures real-time video and uses facial recognition to identify students.

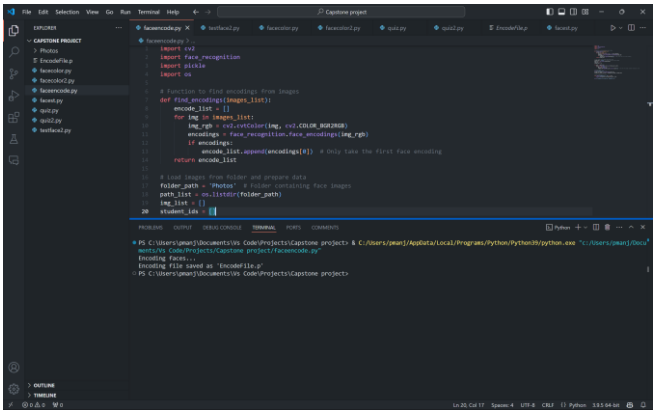


Fig 2: Sample Output of the Encoding File

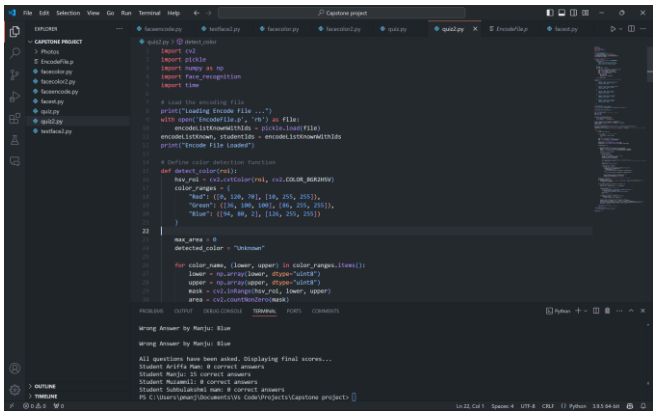


Fig 3: Sample Output the System

It then defines a Region of Interest (ROI) near the student's face to detect colored cards, which students use to answer quiz questions. The ROI is converted from BGR to HSV color space for accurate color detection, leveraging HSV's resilience to lighting variations. Predefined HSV ranges help isolate and identify the colors displayed by the students. The system matches these detected colors with the correct answers for the quiz questions, updating scores in real-time. By using this combination of facial recognition and color detection, the methodology offers an inclusive and accessible way for students to interact with the quiz.

IV. Results and Discussion

To evaluate the effectiveness of SmartSignQuiz, usability testing was carried out with a target group of students facing communication challenges. Metrics such as identification accuracy, response time, and user engagement were monitored. Additionally, the system's impact was compared against conventional quiz methods, highlighting improvements in participation and comprehension. The adaptive color-coded feedback and personalized response tracking demonstrated significant benefits in retaining information, validating the model's effectiveness in fostering an inclusive and engaging learning environment.

The results indicated that the SmartSignQuiz interface was perceived as highly intuitive by all participants. The colour-based design was praised for its user-friendly nature, with participants appreciating the visual clarity provided by the color-coded feedback.

This design facilitated quick comprehension of correct and incorrect responses, aligning well with the visual learning preferences of the users. The real-time feedback mechanism was particularly well-received for its role in maintaining engagement and encouraging continued interaction with the quiz content.

In terms of learning outcomes, participants demonstrated a noticeable improvement in their performance on quizzes after using SmartSignQuiz. The adaptive learning algorithms of the system adjusted question difficulty based on individual progress, which helped reinforce learning objectives and support better retention of material. This personalized feedback proved effective in enhancing students' understanding of the content.

However, some challenges were identified during the testing phase. A few participants experienced minor issues with the facial and gesture recognition components of the system, leading to occasional inaccuracies in interpreting responses. These issues have been noted and will be addressed through further refinements to ensure more reliable performance.

Overall, the initial results from the one-week testing period suggest that SmartSignQuiz is a promising tool for enhancing the educational experience of deaf and hard-of-hearing students. Its innovative approach to colour-based interaction and adaptive feedback has shown potential in improving engagement and learning outcomes. The feedback obtained from this small sample provides a solid foundation for further development and broader testing to ensure the system's reliability and effectiveness.

V. Conclusion

While SmartSignQuiz has shown promising results, several practical limitations need to be addressed. The system's effectiveness is influenced by environmental factors like lighting conditions, which can impact color detection accuracy. Additionally, color differentiation may be

Authorized licensed use limited to: University of New South Wales. Downloaded on October 22,2025 at 13:45:08 UTC from IEEE Xplore. Restrictions apply.

challenging for students with color vision deficiencies, suggesting the need for alternative interaction methods. Hardware limitations, such as camera quality, can also affect facial recognition precision. Future iterations will focus on mitigating these constraints through advanced image processing techniques and adaptable interface design.

The one-week usability testing with a small group of participants demonstrated that the system is highly effective in enhancing student engagement and comprehension. The color-coded feedback and adaptive learning algorithms contributed significantly to improved quiz performance and better retention of material. These features align well with the visual learning preferences of the target audience, providing a clear and immediate understanding of responses.

Despite the overall positive feedback, the testing phase revealed some challenges, particularly concerning the accuracy of the facial and gesture recognition components. These issues underscore the need for further refinement to ensure consistent and reliable performance. Nevertheless, the results indicate that SmartSignQuiz holds significant promise in advancing the field of educational technology for deaf and hard-of-hearing students. Future development and broader testing will be crucial to address the identified challenges and to enhance the system's capabilities. Overall, SmartSignQuiz represents a substantial step forward in creating inclusive educational tools that support diverse learning needs and promote a more equitable learning environment.

VI. Future Enhancements

While the initial deployment of SmartSignQuiz has shown considerable promise, there are several areas for future enhancement that could significantly improve the system's effectiveness and reach. A key focus will be on refining the facial and gesture recognition technologies to enhance accuracy and reliability. This improvement could involve integrating more advanced machine learning algorithms, broadening the range of recognized gestures, and conducting extensive validation across diverse user groups. These steps are essential to ensure that the system can consistently interpret user interactions and provide accurate feedback.

Additionally, expanding SmartSignQuiz to include a wider variety of educational content and interactive features could further enhance its utility. Integrating multimedia elements such as videos and animations, as well as offering customizable quiz formats, would enrich the learning experience. Moreover, incorporating accessibility features for users with color vision deficiencies and actively seeking ongoing feedback from educators and students will help tailor the system to meet diverse needs. Broader testing with larger and more varied populations will also be crucial to validate the system's impact and ensure it effectively supports different educational settings and learning styles.

VII. References

[1] Awotunde, J. B., Misra, S., Florez, H., & Obagwu, O. D. (2024). Multiple Colour Detection of RGB Images using Machine Learning Algorithm. Journal of Computer Science and Technology.

[2] Sainath, J., Reddy, M. S., Saketh, S., Sakshitha, C., Akshay, K., Mohammed, T. K., & Kalyani, A. (2024). Colour Detection using Deep Learning Techniques. International Journal of Artificial Intelligence and Machine Learning.

[3] Nayyer, A., Kumar, A., Rajput, A., Patil, S., Kamat, P., Wagle, S., & Choudhury, T. (2024). Colour-Driven Object Recognition: A Novel Approach Combining Colour Detection and Machine Learning Techniques. Computer Vision and Image Understanding Journal.

[4] Vitthal, P. S., Balasubramanian, S., & Mane, P. S. (2024). Colour Analysis and Classification Based on Machine Learning Technique Using RGB Camera. Textile Engineering and Technology Journal.

[5] Toyokawa, Y., Horikoshi, I., Majumdar, R., & Ogata, H. (2024). Challenges and Opportunities of AI in Inclusive Education: A Case Study of Data-Enhanced Active Reading in Japan. Journal of Inclusive Education and Technology.

[6] Sultan, H., Waris, A., Zafar, M. H., Ijaz, H., Anwer, S., & Sarwar, M. (2024). Real Time Face Recognition Based Attendance System For University Classroom. International Journal of Computer Vision and Pattern Recognition.

[7] Ahmad, I., Sharma, S., Singh, R., Gehlot, A., Gupta, L. R., Thakur, A. K., Priyadarshi, N., & Twala, B. (2024). Inclusive Learning using Industry 4.0 Technologies: Addressing Student Diversity in Modern Education. Journal of Educational Technology and Innovation.

[8] Abdrakhmanov, R., Tuimebayev, A., Zhussipbek, B., Utebayev, K., Nakhipova, V., Alchinbayeva, O., Makhanova, G., & Kazhybayev, O. (2024). Applying Computer Vision and Machine Learning Techniques in STEM-Education Self-Study. STEM Education Journal.

[9] Battina, D. S., & S, L. (2024). Innovative Study of an AI Voice-Based Smart Device to Assist Deaf People in Understanding and Responding to Their Body Language. Journal of Assistive Technologies and Innovations.

[10] Dubey, C. K. (2024). Exploring the Role of Artificial Intelligence in Inclusive Education. SGT University Journal of Education and Technology.

[11] Sun, J.-H., Yang, J.-K., Ji, T.-T., Ji, G.-R., & Zhang, S.-B. (2024). Research on the Hand Gesture Recognition Based on Deep Learning. Proceedings of Computer Vision and AI Applications, Ocean University of China, Qingdao, China.

[12] Kumar, A., Pathak, N., Kirola, M., Sharma, N., Rajakumar, B., & Joshi, K. (2024). AI-Based Mouse Using Face Recognition and Hand Gesture Recognition. Proceedings of Human-Computer Interaction Innovations, U.I.T. Uttaranchal University, Dehradun, India.

Authorized licensed use limited to: University of New South Wales. Downloaded on October 22,2025 at 13:45:08 UTC from IEEE Xplore. Restrictions apply.